

# **(TR-104) PYTHON WITH AI/ML**

## **Training Day 69 Report**

**Date: 16 April 2026**

### **Topic Covered: AI Workflow Builder Enhancement & Video Generation Pipeline**

The sixty-ninth day of training focused on improving the AI Workflow Builder by integrating Instagram Reel generation, video generation pipelines, workflow execution handling, and frontend-backend integration. The training also included exploration of advanced AI video generation models and project documentation setup.

- Designed and implemented an Instagram Reel Configuration input node for collecting structured user inputs.
- Developed dynamic prompt generation logic for AI-based reel script creation.
- Integrated the input node with the AI agent for smooth workflow execution.
- Configured Ollama model for generating Instagram reel scripts.
- Integrated Tavily Search tool for fetching real-time trending content.
- Built a Video Display node for rendering generated videos in the workflow interface.
- Connected frontend workflow with backend video generation pipeline.
- Implemented workflow execution states including idle, running, success, and error handling.
- Debugged and resolved dependency-related issues affecting video generation workflows.
- Improved overall pipeline execution flow from input collection to final video output.
- Integrated text-to-video generation functionality into the workflow system.
- Configured execution support for both CPU and GPU environments.
- Applied optimization techniques for improving video generation performance.
- Managed video generation flow and output handling mechanisms.

- Explored advanced video generation models including Google Veo 3 and Nano Banana.
- Analyzed API requirements and workflow compatibility for advanced video generation systems.
- Identified execution limitations caused by paid API access and billing restrictions.
- Created detailed README documentation explaining project architecture and workflow.
- Documented backend and frontend setup instructions for project execution.
- Organized requirements.txt with all required project dependencies.
- **GitHub Link:**  
<https://github.com/JaspinderKaurWalia26/workflows>
- **Blog Link:**  
<https://medium.com/@jaspinderkaurwalia855/building-an-intelligent-workflow-a6da07a0d381>

Overall, the training strengthened understanding of workflow automation, frontend-backend integration, AI video generation pipelines, advanced video generation models, and project documentation practices.